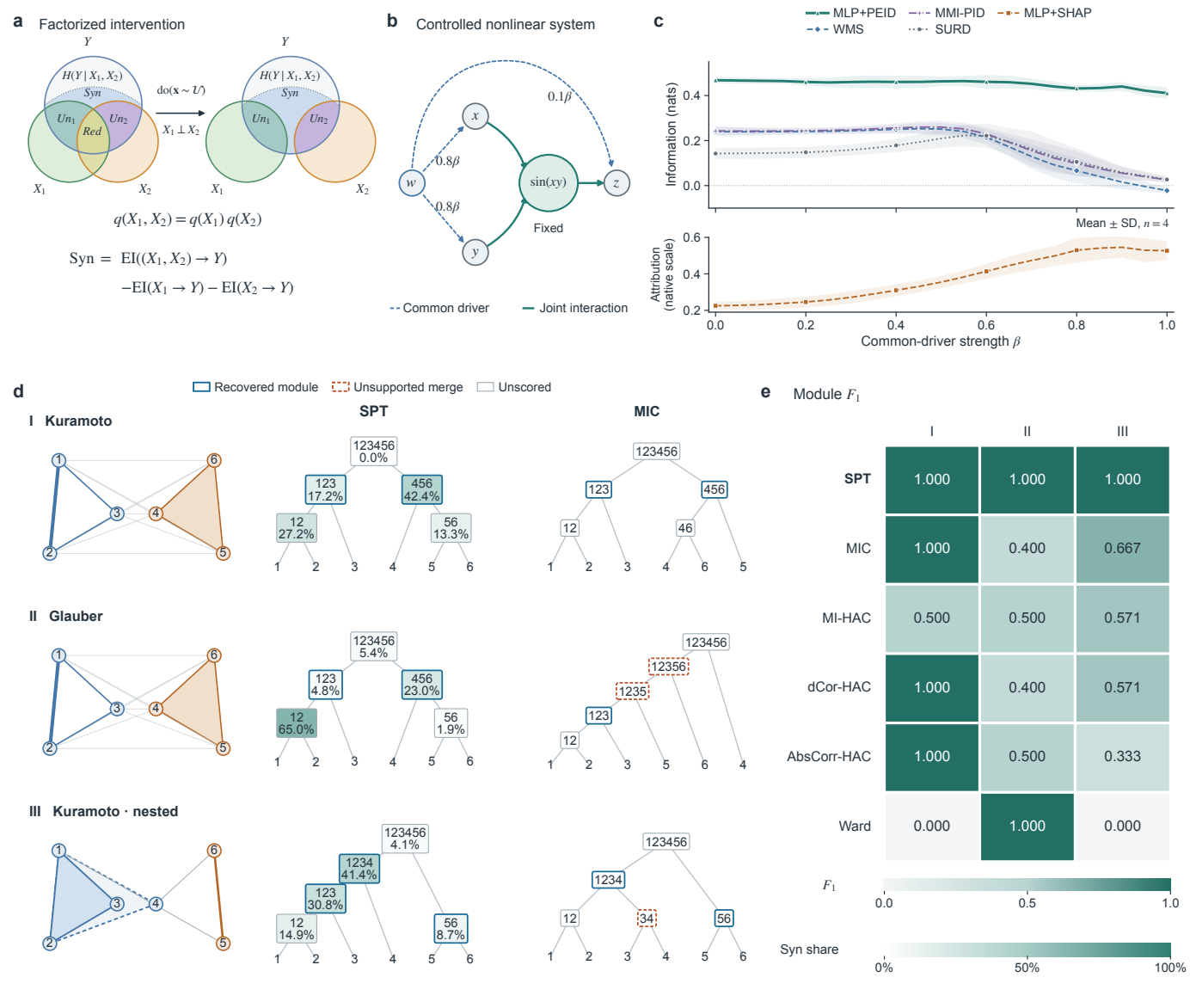

a
Factorized intervention
Y
$H(Y \mid X_1, X_2)$
Syn
Un_1
Un_2
Red
X_1
X_2
$\mathrm{do}(\mathbf{x} \sim \mathcal{U})$
$X_1 \perp X_2$
$q(X_1, X_2) = q(X_1)\, q(X_2)$
$\mathrm{Syn} = \mathrm{EI}((X_1, X_2) \to Y)$
$-\mathrm{EI}(X_1 \to Y) - \mathrm{EI}(X_2 \to Y)$
b
Controlled nonlinear system
0.1β
0.8β
0.8β
x
w
y
z
sin(xy)
Fixed
Common driver
Joint interaction
c
MLP+PEID
MMI-PID
MLP+SHAP
WMS
SURD
Information (nats)
Mean ± SD, $n = 4$
Attribution (native scale)
Common-driver strength β
d
Recovered module
Unsupported merge
Unscored
I Kuramoto
II Glauber
III Kuramoto · nested
SPT
MIC
123456 0.0%
123 17.2%
456 42.4%
12 27.2%
56 13.3%
123456 5.4%
123 4.8%
456 23.0%
12 65.0%
56 1.9%
123456 4.1%
1234 41.4%
123 30.8%
12 14.9%
56 8.7%
123456
123
456
12
46
12356
1235
34
e
Module F_1
I
II
III
SPT 1.000 1.000 1.000
MIC 1.000 0.400 0.667
MI-HAC 0.500 0.500 0.571
dCor-HAC 1.000 0.400 0.571
AbsCorr-HAC 1.000 0.500 0.333
Ward 0.000 1.000 0.000
F_1
Syn share